

Categorical predictors

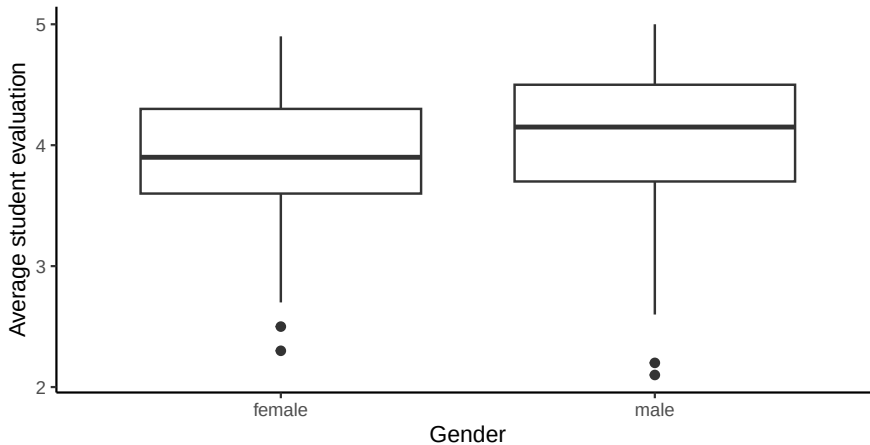
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Categorical predictors with 2 categories

Categorical predictors with 3+ categories

What's the impact of gender on student evaluations?





- Gender is a categorical variable (male or female in this data set) so we can't use it as-is as a predictor
- Idea: recode gender into the quantitative variable 1 = male, 0 = female
- The ordering is totally arbitrary! If you put a categorical variable into a model, R will arbitrarily pick one category or the other as the "1" category, and make the other the "0" category
- The model fit (predictions, R^2 , residual standard error) doesn't change, only the interpretation of coefficients

```
model0 <- lm(eval ~ gender, data=profs)
summary(model0)
```

```
Call:
lm(formula = eval ~ gender, data = profs)
```

```
Residuals:
      Min       1Q   Median       3Q      Max
-1.96903 -0.36903  0.03097  0.43097  0.99897
```

```
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)   3.90103     0.03933   99.19  < 2e-16 ***
gendermale    0.16800     0.05169    3.25  0.00124 **
```

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 0.5492 on 461 degrees of freedom
Multiple R-squared:  0.0224, Adjusted R-squared:  0.02028
F-statistic: 10.56 on 1 and 461 DF,  p-value: 0.001239
```

Compare the fitted model to the means of the two groups:

```
print(model0)
```

Call:

```
lm(formula = eval ~ gender, data = profs)
```

Coefficients:

```
(Intercept)    gendermale  
      3.901         0.168
```

```
profs %>% group_by(gender) %>%  
  summarise(gender_means=mean(eval))
```

```
# A tibble: 2 x 2
```

	gender	gender_means
	<chr>	<dbl>
1	female	3.90
2	male	4.07

What's going on here?



You can pick a reference category using relevel:

```
profs_releveled = profs %>%  
  mutate(gender=relevel(factor(gender), "male"))  
model0_relevel <- lm(eval ~ gender,  
  data=profs_releveled)
```

```
summary(model0_relevel)
```

Call:

```
lm(formula = eval ~ gender, data = profs_releveled)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-1.96903	-0.36903	0.03097	0.43097	0.99897

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	4.06903	0.03355	121.29	< 2e-16 ***
genderfemale	-0.16800	0.05169	-3.25	0.00124 **

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summary(model0)
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```
Call:
```

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```



- We can easily include categorical variables in *multiple* regression models too
- We still need to think carefully about their interpretation, just like other multiple regression coefficients.

```
model1 <- lm(eval ~ beauty + gender, data=profs)
summary(model1)
```

```
Call:
lm(formula = eval ~ beauty + gender, data = profs)
```

```
Residuals:
```

	Min	1Q	Median	3Q	Max
	-1.87196	-0.36913	0.03493	0.39919	1.03237

```
Coefficients:
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	3.88377	0.03866	100.47	< 2e-16 ***
beauty	0.14859	0.03195	4.65	4.34e-06 ***
gendermale	0.19781	0.05098	3.88	0.00012 ***

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 0.5373 on 460 degrees of freedom
```

```
Multiple R-squared:  0.0663, Adjusted R-squared:  0.06224
```

```
F-statistic: 16.33 on 2 and 460 DF,  p-value: 1.407e-07
```

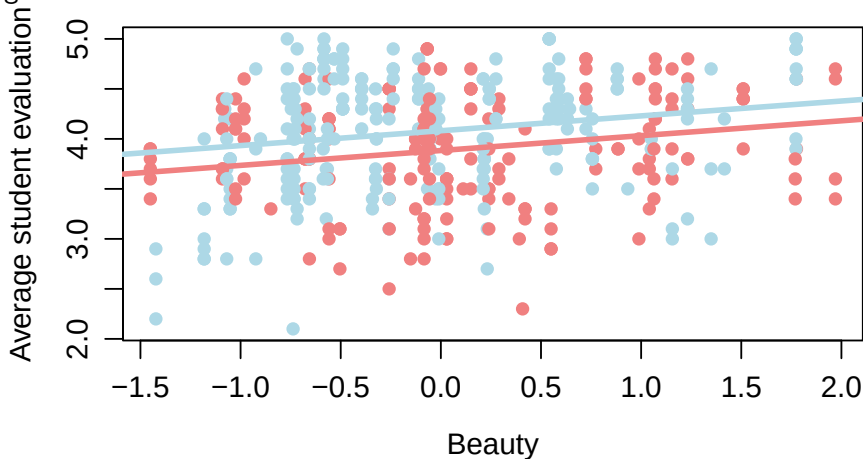


Interpretation:

- Among instructors with the same beauty score, men have predicted average evaluations that are about 0.20 points higher than women
- Among instructors of the same gender, an increase of one standard deviation on the beauty score will lead to an increase in predicted average evaluations of about 0.15 points

Just like last week! The only difference is that an “increase of one unit” in a categorical variable means changing the category

A multiple regression predicting evaluation score from beauty and gender effectively fits two **parallel** regression lines:



Categorical predictors with 2 categories

Categorical predictors with 3+ categories



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- But we can't just create a variable that is 0 for silent generation, 1 for baby boomers, and 2 for gen X—why not?



- The variable `generation` is either `silent` (born before 1945), `boomer` (born 1945-1964), or `genx` (born after 1965)
- Is generation a significant predictor of evaluations **above and beyond** gender and beauty?
- To answer this, we need a model that includes as predictors all of gender, beauty, and `generation`
- But we can't just create a variable that is 0 for silent generation, 1 for baby boomers, and 2 for gen X—why not?
- Solution is to pick a “reference category” and create dummy variables for the **other** categories



Let's arbitrarily pick boomers as a reference category:

Category	genx	silent
Boomers	0	0
Gen Xers	1	0
Silent Gens	0	1

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Category	genx	silent
Boomers	0	0
Gen Xers	1	0
Silent Gens	0	1

R will do this automatically when you add a categorical variable with 3+ categories to a regression (it will arbitrarily pick a reference category)!

```
model2 <- lm(eval ~ beauty + gender + generation, data=profs)
summary(model2)
```

Call:

```
lm(formula = eval ~ beauty + gender + generation, data = profs)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-1.91613	-0.36042	0.03609	0.42282	1.04398

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	3.89727	0.04403	88.521	< 2e-16 ***
beauty	0.14021	0.03304	4.243	2.67e-05 ***
gendermale	0.22230	0.05345	4.159	3.82e-05 ***
generationgenx	-0.02831	0.06149	-0.460	0.6454
generationsilent	-0.16292	0.07992	-2.039	0.0421 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.536 on 458 degrees of freedom

Multiple R-squared: 0.07477, Adjusted R-squared: 0.06669

F-statistic: 9.253 on 4 and 458 DF, p-value: 3.386e-07

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	3.89727	0.04403	88.521	< 2e-16	***
beauty	0.14021	0.03304	4.243	2.67e-05	***
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All else equal (i.e., among professors of the same gender and beauty):

- Gen X professors are predicted to get scores that are 0.03 points below those of boomers

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- Gen X professors are predicted to get scores that are 0.03 points below those of boomers
- Silent gen professors are predicted to get scores that are 0.16 points below those of boomers
- Only the boomer/silent generation difference is statistically significant; gen X professors are not significantly different than boomers
- In other words: age only seems to matter if you are really old